The Role of Oxygen on Anisotropy in Chromium Oxide Hard Mask Etching for Sub-Micron Fabrication

Huseyin Ekinci, Navid M.S. Jahed, Mohammad Soltani, and Bo Cui

***Abstract*—Chromium and its oxides have been playing a vital role in the fabrication of micro- and nano-scale structures in numerous applications for several decades. Controllable, robust and anisotropically dry-etched hard masks and their optimal etch recipes are required in state-of-the-art device fabrication techniques. In terms of manufacturability and repeatability, a mechanistic understanding of the plasma-etching process of chromium oxide (Cr_2O_3) is necessary for its adoption as a hard mask. We present a systematic investigation of plasma etching of chromium oxide films via an inductively coupled plasma-reactive ion etching (ICP-RIE) system in nanoscale. The effects of plasma composition, ICP source power and HF platen power on the etch rate, sidewall profile, surface morphology, and dc-bias have been methodically investigated. We paid particular attention to studying how oxygen content can be used to control the etch profile of nano trenches using chlorine/oxygen gas mixtures, including extremes of very low and very high oxygen content. It was found that chromium oxide etch mechanisms are dependent strongly on the oxygen level. We achieved desirable vertical sidewalls with reasonable etch rates when the oxygen content is in the range 10–40% in the plasma. Oxygen content below 10% resulted in positively tapered etch profiles with low etch rates. On the other hand, bowl-like etch profiles with undercut formation was observed at high oxygen content above 40%, caused by re-emission of the reactive species at this regime. As a hard mask material, patterning Cr_2O_3 films compared to Cr metal is advantageous in terms of etch uniformity and reproducibility. Contrary to Cr, Cr_2O_3 is not as sensitive to chamber wall conditions.**

***Index Terms*—Cl_2/O_2 gas mixture, Cr_2O_3 processing, etch profile, inductively coupled plasma (ICP), nanofabrication, plasma etch, oxygen content.**

I. Introduction

LITHOGRAPHICAL pattern transferring onto underlaying structures can be realized either by the lift-off or direct etching technique with an intermediate hard mask layer. The lift-off technique is very popular in a research context due to its simplicity and low cost. Moreover, it is inevitable in cases where a direct etching of structural material would have undesirable effects on the underlaying layer. However, the lift-off procedure suffers from several drawbacks depending on the type of the resist, sidewall profile, and the method of deposition for the material to be lifted. To achieve reproducible well-defined structures without fence-like features around the pattern edges, the coating material must not be deposited onto the resist sidewalls, which is impossible in isotropic sputter processes. Also, the existing photoresist structures can thermally deteriorate if there is a high heating of the substrate due to the type of coating process and its duration. Moreover, during the liftoff process it is possible that metal flakes in the solution may reattach at undesirable locations on the surface, or unwanted parts of the metal layer may remain on the wafer.

On the other hand, the direct etching pattern transferring method is preferred for high resolution nanofabrication with plasma etching. In this technique, an intermediate hard mask layer is first patterned, and then this layer acts as a hard mask to etch the underlaying layers or substrate. Hard mask material selection and optimizing the etch recipe are critical to achieve desirable surface roughness and sidewall profile. The thickness of the hard mask and etch selectivity (functional material etch rate to mask etch rate) must be enough to leave the mask satisfactorily intact in plasma etching. Moreover, these factors strongly determine the etch selectivity, final structure's morphology, sidewall smoothness, taper angle, and feature resolution [1]–[3].

Over the past few decades, chromium metal (Cr) has been commonly utilized in a large variety of the semiconductor processing fields as a hard mask material in device fabrication [4]–[6], a light-blocking opaque layer in photomask and reticle manufacturing [7]–[9], electrodes for thin film transistor liquid crystal displays and field emission displays [10], [11]. Plasma etching of Cr has been performed in chlorine-based plasmas such as $O_2 + Cl_2$ or $O_2 + CCl_4$ environments, where oxygen and chlorine radicals are responsible for etching by forming the volatile etch by-product CrO_2Cl_2 [5], [12]–[16]. Employing Cr as an intermediate etch mask is challenging due to the necessity of oxygen along with chlorine gas during the plasma etching. However, this causes undesirable results since the polymer resists are known to show poor etch resistance against plasmas including oxygen gas [17].

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Chromium oxide (Cr_2O_3) films are widely utilized in versatile applications such as catalyst supporter [18], green pigment and protective coating [19], gas sensors [20], p-type transparent conducting oxide [21], buffer layers in organic solar cells [22], and anti-reflection layer for photomasks [23]. Moreover, it has been recently reported that chromium oxide is a more suitable hard mask material compared to chrome metal in the case of pattern transferring via plasma etching. Also, both materials have a similar etch resistance and selectivity over silicon in SF_6-based plasmas [24]. Since Cr_2O_3 contains oxygen, it is more readily etched in Cl_2/O_2 plasmas with less oxygen content than is required for Cr metal etching. It is therefore advantageous to transfer the pattern onto Cr_2O_3 rather than metallic Cr even using polymer resists which are easily etched with oxygen in the plasma.

Patterning Cr_2O_3 films acting as a hard mask for the underlaying structure was not systemically investigated in literature. Achieving controllable and anisotropically plasma-etched features is necessary in several applications. In terms of repeatability and manufacturability, mechanistic understanding of the plasma-etching process of chromium oxide is required. In this study, we systematically studied the plasma etching of Cr_2O_3 films in Cl_2/O_2 plasmas. The role of the additive O_2 gas and the relationship between the gas-mixing ratio, and the etch rate, sidewall profile, and surface morphology are discussed in detail at different etch conditions.

II. Experiment

The deposition of the Cr_2O_3 films on 4” single side polished prime silicon wafers was performed using an e-beam evaporator system. The Cr_2O_3 pellets were specified at 99.99% purity. The final thickness of the film was around 500 nm, confirmed by the scanning electron microscopy (SEM) and spectroscopic ellipsometry techniques. The deposition was carried out at room temperature. The hard mask for the Cr_2O_3 etching was a silicon oxide (SiO_2) layer with a thickness of 50 nm, which was deposited by the e-beam evaporator system and patterned with the lift-off technique. Silicon dioxide is known to hold up well in Cl_2 and O_2 chemistry [25]. Also, it allows to characterize the chromium oxide pattern anisotropy independent of mask erosion. However, the ion bombardment must be optimized by keeping the table bias low in order to reduce sputtering of the mask, maximizing selectivity. As low as 1:20 etch selectivity SiO_2:Cr_2O_3 was typically observed in our experiments.

The Cr_2O_3 films were etched using an Oxford PlasmaLab ICP-380 inductively coupled plasma reactive ion etching (ICP-RIE) system, which provides high-density plasma with independently controlled system parameters. The table temperature during the etching was controlled by a thermocouple. Helium backside cooling is used for thermal conduction between the wafer and electrode. The backside helium cooling was maintained at 10 Torr to cool down the chuck system in all our experiments. The gas flows were regulated by mass flow controllers in units of standard cubic centimeters per minute (sccm). The plasma gas chemistry consisted of Cl_2/O_2 mixture, and the total gas flow was fixed to 50 sccm. Etch profile and surface roughness were

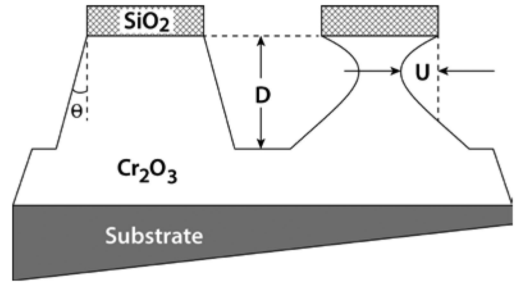


Fig. 1. Schematic of the cross-sectional etch profile of SiO_2/Cr_2O_3 on silicon substrate features. U and D illustrate the undercut and etch depth, respectively. Θ is a measure of positively tapered etch profiles.

investigated under different etch conditions where the system parameters including ICP coil power, HF (high frequency, 13.56 MHz) platen power (also known as RF chuck power), chamber pressure and gas chemistry were modified. While keeping the total gas flow fixed, the ratio between Cl_2 and O_2 was gradually changed to study the effect of O_2 content on the etching parameters. The etch rate and sidewall profile were investigated through cross sectional SEM images. For each etching condition, the sample was cleaved, and cross-sectional SEM images were taken from the trenches in nano scale. An average value was calculated from the images taken from different cross-sections and $\pm 5\%$ error was assumed during the measurements. Fig. 1 represents the measurements of etch profile characteristics where D and U denote as the vertical and lateral etch (undercut) depths, respectively. The undercut, U, was always considered as the deepest part of the lateral etch. Positively tapered etch profile is defined as the sloped etch profile without lateral undercut.

III. Results and Discussion

Etch characteristics depend strictly on plasma gas chemistry, HF platen power, ICP coil source power, chamber pressure, and total flow rate. Typical chlorine-based plasma working at high chamber pressure yield a high Cl ion and neutral concentrations. Thus, this etch regime enhances the chemical etch component of the plasma with a high etch rate, which can lead to round shape etched features and formation of undercut below the hard mask. On the other hand, working at low chamber pressure where the mean free path of plasma species (ions and neutrals) are larger can result in reduction in the etch rate due to the influence of the physical etch component of the plasma. Therefore, in this etch regime it is possible to observe mask erosion that is detrimental for achieving vertical sidewall profiles. Taking into account these points, we set the pressure to 5 mTorr in each set of experiments. Another consideration, RF power that is coupled capacitively through the substrate creates a negative bias (dc-bias) on the electrode with respect to the plasma. By convention, we used their positive values in the figures. High powers generate narrower ion angular distributions that increase directionality [26]. However, highly energetic ion bombardment can cause mask erosion due to sputtering. To minimize this effect on the mask and carrier wafer, we set HF platen power to 20 W in our experiments unless it is stated differently. Since

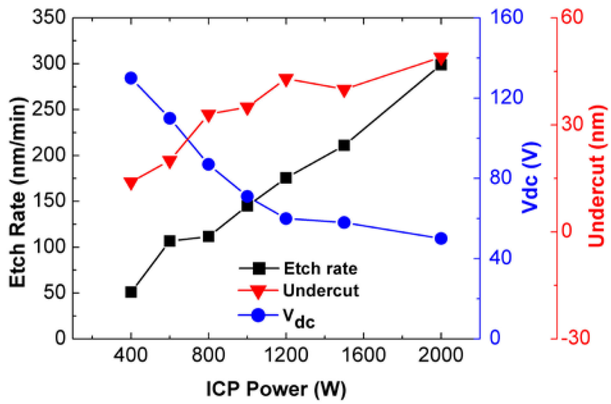


Fig. 2. Cr_2O_3 etch parameters as a function of ICP source power. The effect of ICP power was investigated while keeping the rest of the system parameters constant at 5 mTorr chamber pressure, 20 W HF platen power, and a mixture of chlorine and oxygen in a ratio of 40:10 sccm at 40 °C.

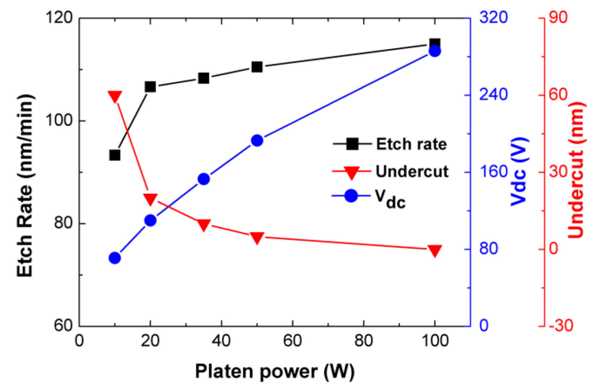


Fig. 3. Cr_2O_3 etch parameters as a function of HF platen power; the rest of the system parameters were set to 5 mTorr of chamber pressure, 600 W of ICP power with a mixture of chlorine and oxygen in a ratio of 40:10 sccm at 40 °C.

the etch by-product CrO_2Cl_2 is volatile at room temperature, we kept the etch temperature at 40 °C in each set of experiments. Since etching in chlorine-based chemistry is very sensitive to the chamber wall condition [27], [28], chamber preconditioning with the etch gas composition was performed for 5 min with a designated silicon wafer, prior to etching actual samples. Also, to ignite the plasma, a strike step was applied with 30W platen power for 5 s prior to main etch step.

In Cl-containing plasma chemistries, the main plasma etch reaction of chromium oxide film was proposed to be

$$\mathrm{CrO_x} + (2-\mathrm{x})\,\mathrm{O}^* + 2\mathrm{Cl}^* \rightarrow \mathrm{CrO_2Cl_2} \tag{1}$$

where O^* and Cl^* are neutral free radicals generated in the plasma [9], [29]. Addition of oxygen to the chamber results in the etch by-product CrO_2Cl_2 (with a boiling point 117 °C), which is volatile at room temperature. Otherwise, no presence of oxygen, possible etch by-products $CrCl_n$ ($n = 1 - 3$) have very low vapor pressures at room temperature (melting point of $CrCl_3$ is 1152 °C, and boiling/decomposition point ~1300 °C) [24]. Therefore, similar to the case of Cr metal etching, the oxygen partial pressure is found to be a dominant parameter in the chromium oxide etching [30].

Firstly, the effect of ICP source power on the etch rate, dc-bias, and undercut was investigated, and the results are shown in Fig. 2. While increasing the ICP power in this set of experiments, the rest of the system parameters were set to 5 mTorr chamber pressure, 20 W HF platen power with a mixture of chlorine and oxygen mixed in a ratio of 40:10 sccm. Working at high ICP powers can produce more reactive ions and neutrals, which consequently results in higher etch rates through the formation of etch products and their faster desorption. Additional Cl-based radicals in the plasma can be generated faster with the increase of ICP power due to their relatively small ionization energies. It is observed that the etch rate linearly increased. It should be noted that there is always undercut at each ICP power, however the lower ICP power resulted in the smaller undercut. The increase in the etch rate with increasing source power is due to the higher concentration of reactive species in the plasma, suggesting a reactant-limited regime. This phenomenon is also supported by the observation of the undercut formation especially at high ICP power. It is seen that the dc-bias of the sample chuck was decreased as the ICP power increased, which may be mainly attributed to energy loss in the ions due to collisions with the plasma species.

Secondly, HF platen power was investigated, and the etch rate, dc-bias, and undercut were illustrated in Fig. 3. The platen power was increased from 10 to 100 W while the rest of the system parameters were set to 5 mTorr of chamber pressure, 600 W of ICP power with a mixture of chlorine and oxygen mixed a ratio of 40:10 in sccm. It is known that RIE power facilitates bond-breaking of the material to be etched and enhances the effectiveness of surface sputtering as well as the removal of etch by-products from the surface. Increasing the table electrode power results in an increase of measured dc-bias voltage at the table. Also, it is expected that surface sputtering can result in an increase in the surface roughness due to physical bombardment of the film surface by ions in the plasma. Even though the etch rate noticeably increased with the increase of HF platen power from 10 to 20W, then it gradually increased to approximately 110 nm/min by further increase at the platen power. It is observed that the dc-bias linearly increased with the increasing HF platen power, from 70 V at 10 W to ~300 V at 100 W leading to more energetic ion bombardment on the film surface. Undercut depth was considerably suppressed and vertical sidewalls can be obtained by adjusting the platen power around 50 W and above, which implies a compromise between chemical and physical etching processes. High powers generate narrower ion angular distributions that increase directionality therefore desirable straight sidewalls maintaining the critical dimension are achieved for patterned features. Fig. 4 represents typical 45-degree tilt SEM images with respect to HF platen power after 3 min etching. The SEM imagery revealed that the etched surface roughness deteriorated due to highly energetic ion bombardment of the films surface with the increasing HF platen power from 10, 20, 50 and 100 W in Fig. 4(a)–d, respectively. Smoother etch surface was observed at low platen power, which may be attributed to that the chemical components of the lateral etch

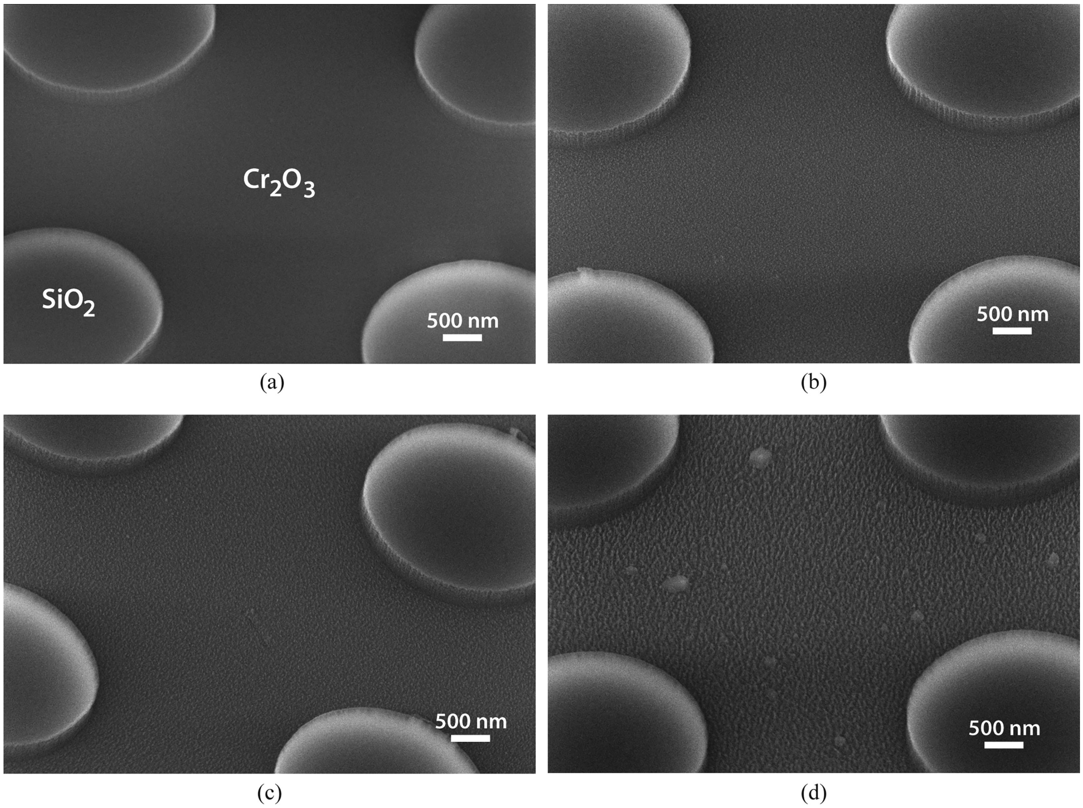


Fig. 4. Tilted (45°) SEM images (after 3 min etching) showing the SiO_2/Cr_2O_3 mesa features as a function of HF platen power at 10, 20, 50, and 100 W represented in (a)–(d), respectively. The etched Cr_2O_3 surface morphology deteriorated due to highly energetic ion bombardment of the film surface with the increasing platen power.

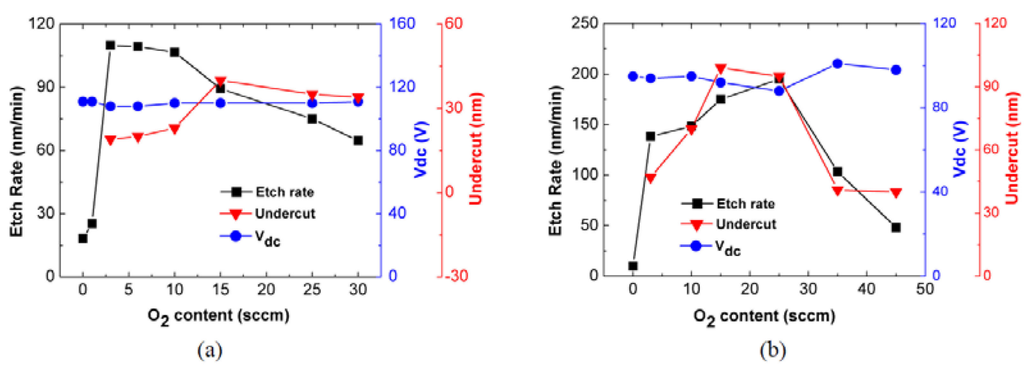
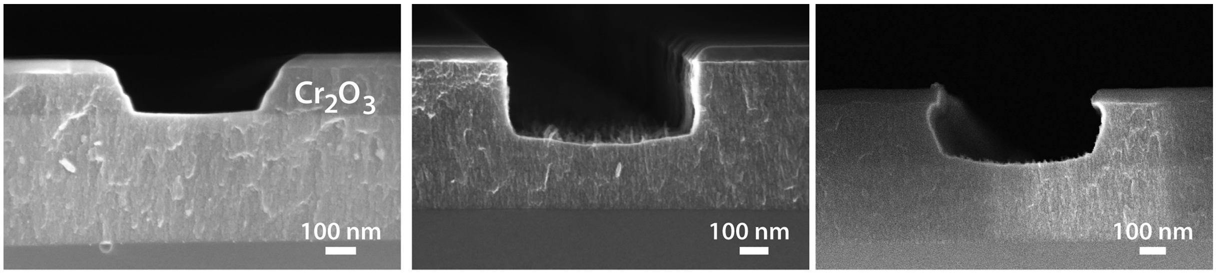


Fig. 5. Cr_2O_3 etch parameters with respect to the oxygen content in the Cl_2/O_2 plasmas at (a) low ICP power (600 W), and at (b) high ICP power (1200 W). The total gas flow was kept constant at 50 sccm for each run.

mechanism dominantly occurred at these system parameters. However, at high HF platen power etch regime the etching is governed by mostly physical bombardment rather than chemical etching.

The effect of oxygen content in the plasma was also studied to investigate the etch characteristics of Cr_2O_3 films. In fact, such investigation was conducted at low ICP condition on Fig. 5(a) and high ICP power condition on Fig. 5(b). It is important to notice that the etch rate was negligibly small without oxygen in the chamber in the case of pure chlorine (50 sccm) in each etch conditions. However, the etch rate is significantly enhanced by introducing oxygen into the chamber even when the oxygen flow rate is as low as 3 sccm. For the low ICP etch condition on Fig. 5(a) where ICP power was set to 600 W (keeping HF platen power at 20 W, 5 mTorr chamber pressure, and a total flow rate of 50 sccm), the etch rate peaked to $\sim$110 nm/min until 10 sccm, then linearly reduced with increasing oxygen content. In this low ICP power regime, it is seen that the etch rate is the highest with an oxygen concentration range of 3–10 sccm (6–20%). Further increase in the oxygen concentration caused to linearly decline the etch rate. This decline was attributed to either a reduction in active chlorine species or the formation of non-volatile reaction products. It is important to note that a considerable amount of oxygen (as high as 50%) is needed to etch chromium metal



(a) (b) (c)

Fig. 6. Cross-sectional SEM images showing the $SiO_2/Cr_2O_3/Si$ etch profiles with respect to the oxygen content in the Cl_2/O_2 plasma while the ICP power, HF platen power, and the chamber pressure were held constant at 600 W, 50 W, and 5 mTorr, respectively. (a) Oxygen content below ∼10 % resulted in positively tapered etch profiles. (b) Vertical sidewalls were achieved in the range of 10–40 % oxygen content with reasonable etch rates and (c) undercuts formed at higher oxygen content.

[13]. However, it is observed that the undercut kept increasing slightly. We surmise that the undercut formation is likely due to the transportation of the neutral species by re-emission from the trench bottom to the sidewalls when an etching precursor desorbs before the reaction toward the final volatile product. Also, reactant sticking coefficients could play a role in the high-oxygen regime. Similar etch profiles have been observed in oxygen containing plasmas in literature to etch chromium and silicon [15], [31], [32]. On the other hand, etching conducted with oxygen content below 3 sccm resulted in a positively tapered etch profile, similar to the one seen in Fig. 6(a). The dc-self bias voltage is nearly constant at ∼110 V throughout the different oxygen/chlorine ratio under investigation. For oxygen content higher than 30 sccm, the plasma was not be ignited, with no measurable dc-bias.

As seen in Fig. 5-(b), in the case of high ICP region where ICP power was set to 1200 W and keeping HF platen power at 20 W, chamber pressure at 12 mTorr with a total flow rate of 50 sccm, the etch rate peaked as high as 200 nm/min at an oxygen content of ∼25 sccm, then drastically reduced with the increasing oxygen content. It was observed that undercut depth was also very large in this etch regime. Even though the oxygen content is very high (O_2:Cl_2 45:5 sccm) there was still etching but with very low rates. It should be noted that the dc-self bias voltage is almost constant ∼95 V regardless of oxygen content in the plasma.

Lastly, the effect of oxygen content at a high platen power was further studied to investigate the etch characteristics of Cr_2O_3 films. In this set of experiments, while changing the chlorine/oxygen mixture ratio in the chamber the ICP power, HF platen power, and the chamber pressure were held constant at 600 W, 50 W, and 5 mTorr, respectively. The total gas flow was fixed to 50 sccm. With respect to oxygen content in the plasma, there were three etch profiles observed from the typical cross-sectional SEM images as seen in Fig. 6. Oxygen content below ∼10% resulted in positively tapered etch profiles as seen in Fig. 6(a).

$CrCl_2O_2$ is the volatile reaction product in both chromium and its oxide etching processes. There are numerous considerations during etch product formations such as the flux of ions and neutrals, sticking coefficients of chlorine and oxygen, rates of surface diffusion, and rates of reaction of volatile productions. For the case of chromium etching process, there are several possible non-volatile intermediates including chromium oxides, Cr_xO_y, chromium chlorides, Cr_xCl_y, or some chromyl chlorides, $Cr_xO_yCl_z$ (e.g., CrOCl is of low volatility) [15]. In this low oxygen concentration regime, away from the optimum gas mixture, it is likely to form non-volatile etch products. With the help of ion bombardment these non-or-low volatile products can be desorbed from the trench bottom and re-deposited on the sidewalls by acting as inhibitors. As a result, positively tampered etch profile is formed

Straight sidewalls as shown in Fig. 6(b) were achieved until the oxygen content reached 40% in the chamber with higher etch rates. This implies that these system parameters resulted in an ion-assisted chemical reaction regime where both chemical and physical etch mechanisms were in place, resulting in anisotropic etch profiles with vertical sidewalls. However, the etch rate decreased when the oxygen content was higher than 40%. Further increasing the oxygen content caused a distortion in the verticality of the sidewalls and undercut formation was observed as shown in Fig. 6(c).

The optimum gas flow depends on the ability to form free radicals (depends on the dissociation energy of O_2 and Cl_2) in the plasma and the sticking coefficient (probability of reaction when hitting the surface). Away from the optimum gas mixture, in the excessively high oxygen concentration in the chamber results in low etch rates due to a reduction in the active chlorine species. This also may be attributed to the weakening anisotropy of the ions by the platen electrode, resulting in more isotropic etching. The undercut formation could likely arise from the poor sticking coefficients and the transportation of the neutral species by re-emission from the trench bottom to the sidewalls as reported earlier [15], [31], [32]. Also, it is possible to see a bowl-shaped bottom etch profile at this etch regime.

IV. Conclusion

In this investigation, the plasma etching of chromium oxide films in nanoscale by an inductively coupled plasma-reactive ion etching (ICP-RIE) system was studied. Working at high ICP powers resulted in higher etch rates but undesirable etch profiles with undercut, whereas low ICP power suppressed the undercut formation. There were three etch profiles observed as a function of oxygen content at high HF platen power regime. Oxygen content below 10% resulted in positively tapered etch profiles with low etch rates. Desirable vertical sidewalls with reasonable etch rates were achieved with oxygen content in the range of 10–40% in the plasma. Finally, low etch rates and bowl-like profiles with undercut formation was observed when the oxygen content was higher than 40%. As a hard mask material, patterning Cr_2O_3 films compared to Cr metal is advantageous in terms of etch uniformity and reproducibility. Since metallic Cr is very sensitive to fluorocarbon polymer in an etch chamber and any such polymer that lands on the surface seems to block the etching, whereas Cr_2O_3 is not as sensitive to chamber wall conditions.

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